University of Ain Temouchent - Department of Computer Science
Field: Computer Science
Module: Natural Language Processing

Second Practical Session: Stemming and Lemmatization

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Stemming and Lemmatization: Practical Assignment

In this notebook, you will learn the basics of stemming and lemmatization using the NLTK library. You will work with the Porter Stemmer for stemming and the WordNet Lemmatizer for lemmatization. You will also explore how to improve lemmatization results using parts-of-speech (POS) tagging.

Objectives

- Understand the difference between stemming and lemmatization.
- Use NLTK's `PorterStemmer` to perform stemming on various words.
- Use NLTK's `WordNetLemmatizer` to perform lemmatization, and see how specifying the POS tag improves results.

Required Libraries

To run this practical session successfully, you need to install the following Python libraries.

Installation

Use the following command in your terminal or Jupyter Notebook:

```
pip install nltk
```

Additionally, since the notebook requires some **NLTK datasets**, download them within your Python script or Jupyter Notebook:

```
import nltk
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
nltk.download('punkt')
```

Step 1: Importing Libraries

First, we need to import the necessary modules from NLTK.
Cell [1]:

```
import nltk
from nltk.stem import PorterStemmer
from nltk.stem import WordNetLemmatizer
from nltk.corpus import wordnet

# Download required NLTK data
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger_eng')
```

Step 2: Basic Stemming with PorterStemmer

We will use the `PorterStemmer` to stem various words. Notice that some words like "walking" are reduced to "walk" while others remain almost unchanged.

```
porter = PorterStemmer()
```

```
print(porter.stem("walking"))
```

Using the `PorterStemmer`, write Python code to stem the following words:

- `"walked"`
- `"walks"`
- `"ran"`
- `"running"`
- `"bosses"`
- `"replacement"`

Step 3: Processing a Sentence with Stemming

Let's see how stemming works on a full sentence by splitting the sentence into tokens and applying the stemmer.

```
sentence = "Lemmatization is more sophisticated than stemming".split()
for token in sentence:
    # complete the code
```

```
sentence = "unnecessary berry".split()
for token in sentence:
    # complete the code
print()
```

Step 4: Basic Lemmatization with WordNetLemmatizer

The `WordNetLemmatizer` is more sophisticated than stemming because it uses a dictionary and requires the correct part-of-speech (POS) tag for best results.

```
lemmatizer = WordNetLemmatizer()
```

```
print(lemmatizer.lemmatize("as"))
```

Using the `WordNetLemmatizer`, write Python code to lemmatize the following words:

- `"walking"`
- `"walking"` (Hint: Use `pos=wordnet.VERB`)
- `"going"`
- `"going"` (Hint: Use `pos=wordnet.VERB`)
- `"ran"` (Hint: Use `pos=wordnet.VERB`)

Comparison with Stemming for Special Cases

In this step, you will compare the results of **stemming** (using the Porter stemmer) and **lemmatization** (using the WordNet lemmatizer) for different words.

Instructions

1. Use the **Porter stemmer** to stem each word.
2. Use the **WordNet lemmatizer** to lemmatize each word.
3. For some words, you will need to specify the **part of speech (pos)** to get the correct lemmatized result.

Words to Compare

- `"mice"`
- `"was"` (Hint: Use `pos=wordnet.VERB` for lemmatization)
- `"is"` (Hint: Use `pos=wordnet.VERB` for lemmatization)
- `"better"` (Hint: Use `pos=wordnet.ADJ` for lemmatization)

Question

- What differences do you observe between stemming and lemmatization?

Step 5: Using POS Tagging to Improve Lemmatization

For lemmatization, providing the correct part-of-speech (POS) tag helps in returning the true base form of a word. First, we define a helper function to convert POS tags from the Penn Treebank format to the format used by WordNet.

```
def get_wordnet_pos(treebank_tag):
    if treebank_tag.startswith('J'):
        return wordnet.ADJ
    elif treebank_tag.startswith('V'):
        return wordnet.VERB
    elif treebank_tag.startswith('N'):
        return wordnet.NOUN
    elif treebank_tag.startswith('R'):
        return wordnet.ADV
    else:
        return wordnet.NOUN
```

```
# Example sentence for POS tagging
sentence = "Donald Trump has a devoted following".split()
words_and_tags = nltk.pos_tag(sentence)
print(words_and_tags)

for word, tag in words_and_tags:
    lemma = # Complete this
    print(lemma, end=" ")
print()
```

Using the same approach as in the previous example, apply **lemmatization with POS tagging** to the following sentence:

Sentence

```
"The cat was following the bird as it flew by"
```